Conflict and Peace

Week 2: Economic Incentives behind Conflicts

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

September 12th, 2025

Stylized facts about conflicts (Blattman, Miguel, 2010)

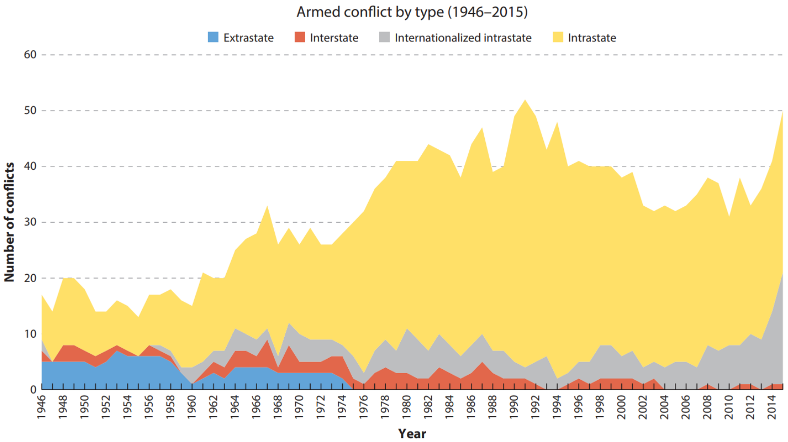
Many conflicts take form of internal conflicts

- Roughly 20-30% of countries have active civil war/conflicts during 1960-2000s
- At a given year: Kill approx 270,000 ppl & 8.44 mil disability-adjusted life-years $\Downarrow$
- Slows down GDP growth (2% yr) & human capital, infrastructure, and institutions $\Downarrow$
- Disproportionately affect poorer regions

Economic roots of (civil) wars

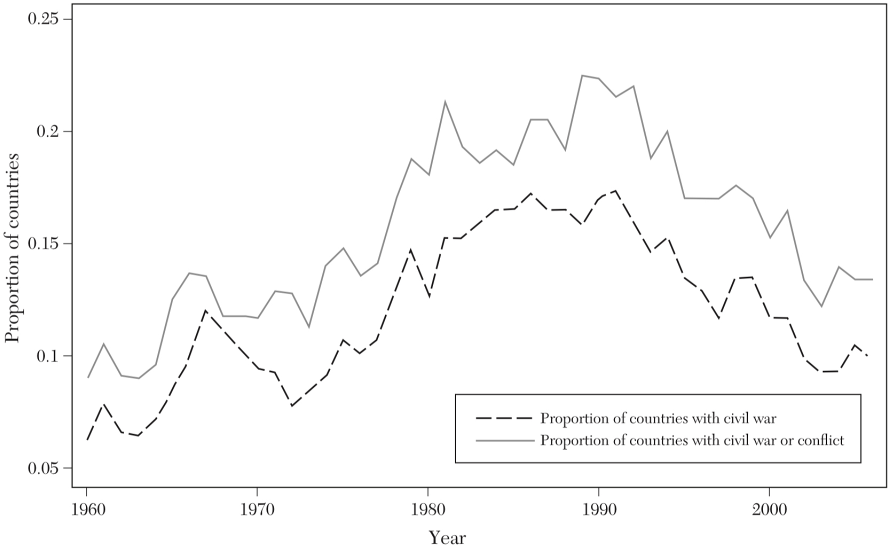
- Poverty and weak economic growth correlated with conflict onset
- Resource abundance raises risk of conflicts via higher rents to be seized
- Individuals more inclined to join conflict when returns to work and peace are low
- Weak state capacity to contain conflict leads to longer duration

Share of wars of different types

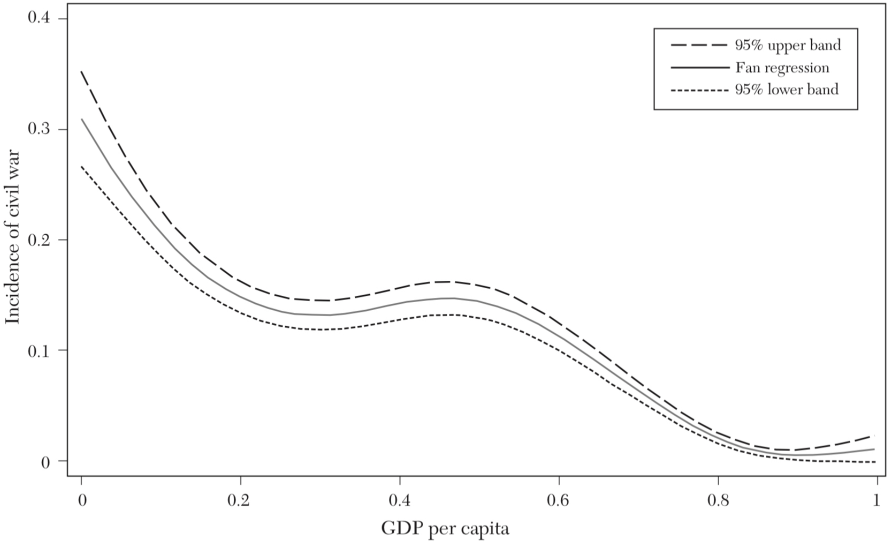


Source: Ray, Esteban (2017) “Conflict and Development", *Annual Reviews of Economics*,9,263-293

Share of countries with active conflicts each year



Disproportionate distribution of conflicts



Why do people fight? (Blattman, 2022)

War is costly, yet people fight due to

- **Unchecked interests:** Decisionmakers that benefit from wars not held accountable
- **Intangible interests:** Preference for values such as identity-based vengeance and glory
- **Uncertainty:** Lack of information on others' (and own) benefits and constraints
- **Commitment Problem:** Inability to trust promises to constrain power over time
- **Misperceptions:** Mistaken beliefs that lead to errors in judgement

These are not mutually exclusive

In all, economic calculuus explains why conflicts occur

We explore how economics and political science approaches the causes of conflict with these baseline reasons in mind

Why? Still an ongoing problem

(a) Israel-Palestine War

(b) Russia-Ukraine War

(c) Religious conflict in Burkina Faso

(d) Martial law attempt (South Korea)

Roadmap for today

1. Overview of the topic

2. Resource and commodity shocks affect conflicts
   - 2.1 Caselli, Morelli, Rohner (2015): The Geography of Interstate Resource Wars
   - 2.2 Dube, Vargas (2013): Comodity Price Shocks and Civil Conflict: Evidence from Colombia
   - 2.3 McGuirk, Burke (2020): The Economic Origins of Conflict in Africa
   - 2.4 Cao, Chen (2022): Rebel on the Canal: Disrupted Trade Access and Social Conflict in China, 1650-1911 [student presentation]

3. Economic opportunities to individuals explain conflicts
   - 3.1 Shapiro, Vanden Eynde (2023): Fiscal Incentives for Conflict: Evidence from India's Red Corridor
   - 3.2 Blattman, Annan (2016): Can Employment Reduce Lawlessness and Rebellion?
   - 3.3 Berman et al (2011): Do Working Men Rebel? Insurgency and Unemployment in Afghanistan, Iraq, and the Philippines [student presentation]

Caselli, Morelli, Rohner (2015): The Geography of Interstate Resource Wars

Q: Resource endowment & Location $\Rightarrow$ Interstate conflict

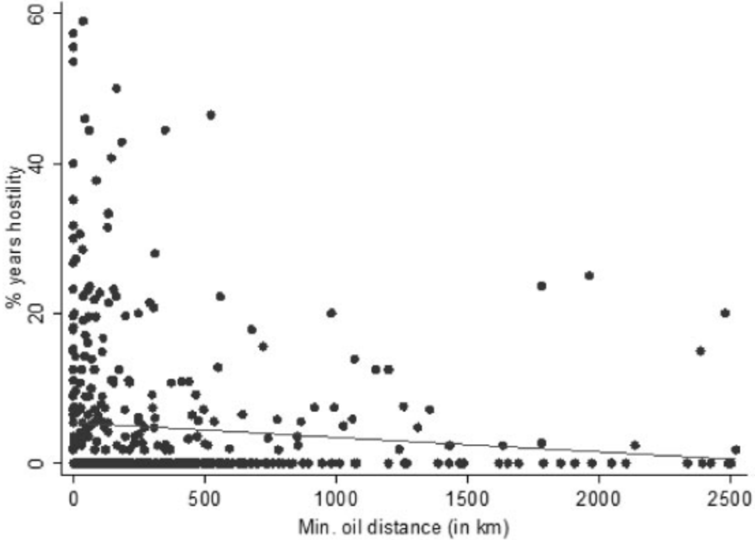
Natural resources as triggers of wars

- Many individual cases studies, but lack systematic causal estimation
- Theoretical predictions
  - Resource endowments closer to the border $\Rightarrow \Uparrow$ Interstate Conflicts
  - The farther the resources, more costlier to claim them
- Empirical analysis
  - Negative correlations between distance and conflicts documented
  - Possible omitted variable bias: Resource-endowed countries innately belligerent?

Key innovation and takeaways

- Incorporate likelihood of conflicts over geographical setting with asymmetrical payoff
- Model how motives of countries respond to resource values and Acquisition costs

Correlation between Conflict and distance to resources



Setting up theoretical framework

Setup for the Model (payoffs in the matrix)

	Country B: Peace	Country B: Conflict
Country A: Peace	$0, 0$	$x + c_A, -x + c_B$
Country A: Conflict	$x + c_A, -x + c_B$	$x + c_A, -x + c_B$

- x is the payoff of conflict to A, $-x$ is the loss from conflict to B (asymmetry)
  - If $x > 0$: A is the expected winner
- c is the net benefit (benefit $-$ cost) of conflict
  - Usually $c < 0$: Conflicts are very costly..!
- Peace persists if
  - for A: $0 \geq x + c_A$ and for B: $0 \geq -x + c_B \Rrightarrow c_B \leq x \leq -c_A$
  - i.e. Conflicts are costly (c_A and c_B far below zero) and gains from conflicts (x) is small

Takeaways from the framework

Testible predictions

- Higher asymmetric payoff ($x \Uparrow$) $\Rrightarrow$ Conflict rises
  - One country has oil > None have oil
  - Two countries have oil > None have oil
  - One country has oil > Two countries have oil (smaller asymmetry in the latter)
- Conflict inversely related with cost of acquisition (distance)
  - One country has oil: Conflict rises when oil is close
  - Both have oil: Changes in distance that increses oil presence asymmetry increase conflicts
- Overall: Payoff asymmetry and lesser cost of acquisition makes conflicts likely

Data and Identification

Sources of data

- Dyadic Militarized Interstate Dispute dataset with 0-5 scale for conflict intensity
- Explanatory variables
  - Georeferenced oil data (PETRODATA) + Distance from borders from CShapes dataset
  - Control: land area, GDP per capita, population, democracy, trade, among others

Regression Approach (by country pair (d)-year (t))

$$\begin{aligned} Y_{d,t+1} = \alpha &+ \beta \text{One}_{d,t} + \gamma \text{One} \times \text{Dist}_{d,t} \\ &+ \delta \text{Both}_{d,t} + \eta \text{Both} \times \text{MinDist}_{d,t} + \text{Both} \times \text{MaxDist}_{d,t} + \xi X_{d,t} + u_{d,t} \end{aligned}$$

- (None/One/Both) countries in pair d has resources
- Control for distance from border (MinDist: Smaller of the distances where Both $= 1$)
- Comparison relative to country pairs without any oils (None)

Key Results

TABLE I

BASELINE RESULTS FOR HOSTILITY

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
One	0.034	0.029	0.049*	0.077**	0.087	0.081**	0.143***	0.124***	0.048	0.063*	0.058*	0.132***
	(0.032)	(0.027)	(0.027)	(0.030)	(0.054)	(0.040)	(0.048)	(0.035)	(0.042)	(0.038)	(0.033)	(0.044)
One × Dist	−0.050	−0.044	−0.073***	−0.086***	−0.107*	−0.087**	−0.138***	−0.118***	−0.079*	−0.072*	−0.103***	−0.137***
	(0.035)	(0.027)	(0.026)	(0.027)	(0.056)	(0.043)	(0.048)	(0.034)	(0.044)	(0.039)	(0.033)	(0.041)
Both	0.022	0.028	0.034	0.045*	0.023	0.048*	0.110***	0.073**	0.009	0.024	0.020	0.047
	(0.021)	(0.020)	(0.029)	(0.027)	(0.030)	(0.027)	(0.035)	(0.030)	(0.024)	(0.020)	(0.032)	(0.031)
Both × MinDist	−0.077**	−0.044	−0.105***	−0.089***	−0.088*	−0.028	−0.107**	−0.066**	−0.102***	−0.096**	−0.122***	−0.125***
	(0.035)	(0.035)	(0.030)	(0.029)	(0.047)	(0.035)	(0.051)	(0.032)	(0.038)	(0.043)	(0.032)	(0.036)
Both × MaxDist	0.026	−0.014	0.016	0.004	0.048	−0.048	0.012	−0.010	0.059	0.042	0.047	0.043
	(0.040)	(0.036)	(0.030)	(0.029)	(0.065)	(0.050)	(0.065)	(0.044)	(0.043)	(0.045)	(0.034)	(0.038)
Type oil	All	All	All	All	Offshore	Offshore	Offshore	Offshore	Onshore	Onshore	Onshore	Onshore
Country FE	No	No	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes
Add. controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Observations	19,962	11,303	19,962	11,303	19,962	11,303	19,962	11,303	19,962	11,303	19,962	11,303
R-squared	0.019	0.090	0.145	0.158	0.020	0.091	0.145	0.156	0.021	0.092	0.146	0.160

Notes. Dependent variable: Hostility. The unit of observation is a country pair in a given year. The sample covers all contiguous country pairs and the years 1946–2001. Method: OLS with robust standard errors clustered at the country-pair level. All explanatory variables are taken as first lag. All specifications control for the average and the absolute difference of land areas in the pair, intercept, and annual time dummies. Additional controls are: The average and absolute difference of GDP per capita, the average and absolute difference of population, the average and absolute difference of fighting capabilities, the average and absolute difference of democracy scores, dummy for one country having civil war, dummy for both countries having civil war, bilateral trade/GDP, dummy for one country being OPEC member, dummy for both countries being OPEC member, genetic distance between the populations of the two countries, dummy for membership in the same defensive alliance, dummy for historical inclusion in the same country, kingdom, or empire, dummy for having been in a colonial relationship, and years since the last hostility in the country pair. Significance levels *** $p < .01$, ** $p < .05$, * $p < .1$.

- When all are controlled, Oil presence $\rightarrow \Uparrow$ Conflcit and $\Uparrow$ Dist $\rightarrow \Downarrow$ Conflict

Mechanism and discussion

Findings are robust to

- Different measures of conflicts (Full scale war only, disupte intensity)
- Controlling for oil production/reserve amounts
- Estimation methods (Logit, Probit regressions)
- Reducing sample to cases where there were no border changes

Key takeaways from the paper

- Asymmetrical oil presence $\rightarrow \Uparrow$ Conflicts
- Distance matters: $\Uparrow$ Distance to oil $\rightarrow \Downarrow$ Conflict
- Most dangerous: Only one country has oil, and it is close
- Thus, conflicts increase when gains are asymmetric and costs of acquisition are low

Dube, Vargas (2013):
Comodity Price Shocks and Civil Conflict:
Evidence from Colombia

Q: How do different income shocks affect conflicts?

Relationship between income and conflict: Complicated

- Reduce conflicts: Wages $\Uparrow$ & Labor supplied to conflict $\Downarrow$ (opporunity cost effect)
- Increase conflcits: Spoils to fight over $\Uparrow$ (rapacity channel)
  - Presumably, different types of income shocks lead to different effect on conflicts

This paper leverages different sources of income shocks to answer this dilemma

- Colombia: Long history of civil conflicts, with high escalation in 1990s
- Agricultural goods (labor-intensivity $\Uparrow$) vs Natural resource prices (labor-intensivity $\Downarrow$)
  - Coffee vs Oil: Colombia's biggest exports
- Use variation in municipal conflict incidence and commodity distribution in Colombia

History of civil conflicts in Colombia

Long history of civil wars

- Left-wing guerrilla (FARC, ELN) vs govt vs right-wing paramilitaries (Oficina Cartel etc)
- Rooted in *La Violencia* in 1948, current form took place in 1960s
- Heightened in 1990s, ongoing low-intensity fights despite peace agreement in 2016

What did they fight over?

- Appropriate resource returns, government revenues through various channels
- Coffee returns huge: One of the largest exporters in the world
- Actively recruited rural workers by paying wages over minimum wage

Data and empirical strategy

Sources of (some of the) data

- Conflict Analysis Resource Center (CERAC) for civil war episodes: 1988-2005
- Commodity distribution: Agricultural land area, oil/gold production, coca production
- Price: International Coffee Organization and Internatinal Financial Statistics

Empirical strategy

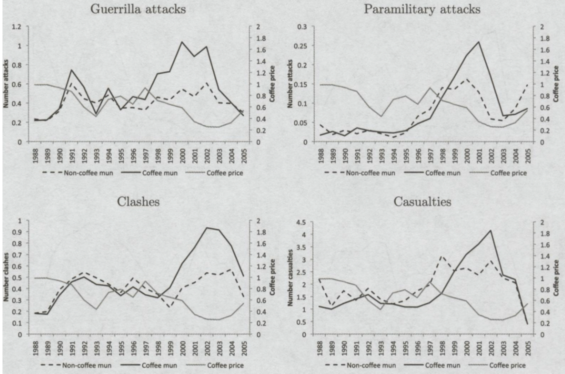
- DID with IV for coffee prices: Colombian conflicts $\rightarrow$ prices directly (reverse causality)

$$Y_{jrt} = \alpha_j + \beta_t + \delta_r t + \gamma \text{Coca}_{jr} t + \lambda(\text{Oil}_{jr} \times PO_t) + \rho(\widehat{\text{Coffee}_{jr} \times PC_t}) + \phi X_{jrt} + e_{jrt}$$

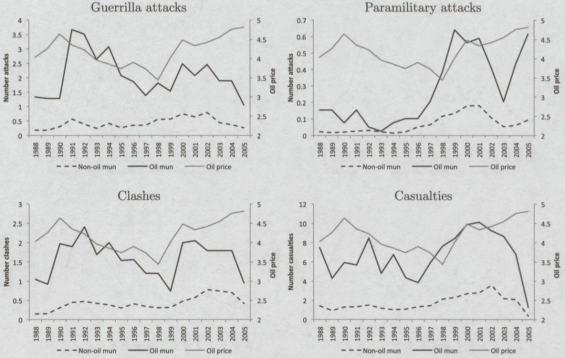
- Rainfall $\times$ Temperature $\times$ Exports of other big producers as IV

$$\text{Coffee}_{jr} \times PC_t = \alpha_j + \beta_t + \delta_r t + \gamma \text{Coca}_{jr} t + \lambda(\text{Oil}_{jr} \times PO_t) + \sum_{m=0}^{1} \sum_{n=0}^{1} \theta_{mn}(R_{jr}^m \times T_{jr}^n \times E_t) + \phi X_{jrt} + e_{jrt}$$

Descriptive findings



(a) Coffee price & trend of violence in coffee vs non-coffee municipalities



(b) Oil price & trend of violence in oil vs non-oil municipalities

Overall effect on violence

TABLE 2

The effect of the coffee and oil shocks on violence

Dependent variables	(1) Guerrilla attacks	(2) Paramilitary attacks	(3) Clashes	(4) Casualties
Coffee int. x log coffee price	−0.611**	−0.160***	−0.712***	−1.828*
	(0.249)	(0.061)	(0.246)	(0.987)
Oil production x log oil price	0.700	0.726***	0.304	1.526
	(1.356)	(0.156)	(0.663)	(2.127)
Observations	17,604	17,604	17,604	17,604

Notes: Standard errors clustered at the department level are shown in parentheses. Variables not shown include municipality fixed effects, year fixed effects, log of population, and linear trends by region and municipalities cultivating coca in 1994. The interaction of the internal coffee price with coffee intensity is instrumented by the interaction of the coffee export volume of Brazil, Vietnam, and Indonesia with rainfall, temperature, and the product of rainfall and temperature.
*** is significant at the 1% level; ** is significant at the 5% level; * is significant at the 10% level

Leading mechanism behind the outcomes

TABLE 3

The opportunity cost and rapacity mechanisms

	(1)	(2)	(3)	(4)	(5)
	Opportunity cost mechanism			Rapacity mechanism	
Dependent variables	Log wage	Log hours	Log capital revenue	Paramilitary political kidnappings	Guerrilla political kidnappings
Coffee int. x log coffee price	0.371*	0.286**	−0.787	0.022	−0.060
	(0.217)	(0.125)	(0.698)	(0.014)	(0.060)
Oil production x log oil price	1.230	0.079	0.419**	0.168***	−0.066
	(0.894)	(0.314)	(0.203)	(0.009)	(0.206)
Observations	26,050	57,743	11,559	16,626	16,626
Sample period	1998–2005	1998–2005	1988–2005	1988–2004	1988–2004

Notes: Standard errors clustered at the department level are shown in parentheses. In column (1), the dependent variable is the log of hourly wage, defined as the the individuals' earnings in the past month divided by hours of employment in the past month. In column (2), log hours refers to hours of employment during the past month. Variables not shown in all specifications include municipality fixed effects, year fixed effects, and linear trends by region and municipalities cultivating coca in 1994. Columns (1) and (2) also control for education, age, age squared, and indicators of gender and marital status. Columns (3)–(5) additionally control for log population. The interaction of the internal coffee price with coffee intensity is instrumented by the interaction of the coffee export volume of Brazil, Vietnam, and Indonesia with rainfall, temperature, and the product of rainfall and temperature.

*** is significant at the 1% level; ** is significant at the 5% level; * is significant at the 10% level

Discussion

Robustness of the outcomes and ruling out other mechanisms

- Results consistent when substitute commodities are used
  - Oil $\rightarrow$ Coal and Gold
- Not driven by changes in population composition following conflict-induced migration
- Not affected by decreasing government presence or enforcement (increases instead)
- Results unaffected when coca-production is accounted for

Key takeaways from the paper

- Different commodity price shocks trigger different channels of conflict
- Labor-intensive price $\Uparrow$ $\rightarrow$ Opportunity cost of forgoing labor $\Uparrow$ $\rightarrow$ Conflict $\Downarrow$
- Less labor-intensive price $\Uparrow$ $\rightarrow$ Maximizable rents $\Uparrow$ $\rightarrow$ Conflict $\Uparrow$

McGuirk, Burke (2020):
The Economic Origins of Conflict in Africa

Q: How do price shocks differently affect producers & consumers?

Explaining how economic shocks can affect conflict is tricky

- Shocks to opportunity cost also affect spoils and capacity to repel insurgents
- Income may affect conflict, and vice versa (reverse causality)
- Both income and conflict may be affected by omitted factors (legal institutions)

This paper differentiates incidence of conflicts by actors and motives

- Leverage food price shocks and differences in food-production across households
  - Grain price $\Uparrow \rightarrow$ Income $\Uparrow$ for producers ($\because$ Higher returns on food production)
  - Grain price $\Uparrow \rightarrow$ Real Income $\Downarrow$ for consuming households ($\because$ Spend more on food)
- Conflict may take different forms
  - Conflict over land, workers, and resources (factor conflict) - aim to displace population
  - Conflict over splitting spoils (output conflict) - more transient

Theoretical framework

Setting

- Consider an economy of producers (landownders) and consumers (work or fight)
- Food price changes affect returns from land & relative wages between work vs fighting

Factor conflict

- Producer price $\Downarrow \rightarrow$ Farming returns $\Downarrow$. Consumper prices $\Uparrow \rightarrow$ Income $\Downarrow$ & Fight $\Uparrow$
- Crop price $\Uparrow$: Reduce factor conflict for producers ($\because$ choose farming over attacking)
- Crop price $\Uparrow$: Increase conflict for consumers ($\because$ choose high-paying armed groups)

Output conflict

- Both consumers and producers can appropriate returns from food price increase
- With higher spoils available, Crop price $\Uparrow \rightarrow$ Conflict $\Uparrow$ for both

Data and Empirical Strategy

Data sources at the grid level data

- Conflict: UCDP, ACLED, and Afrobarometer
  - Factor conflict: involve more organized groups using UCDP
  - Output conflict: riots and violence against civilians using ACLED and Afrobarometer
- Food/consumer price: Spatial distribution of crops + annual world price (UNFAO)

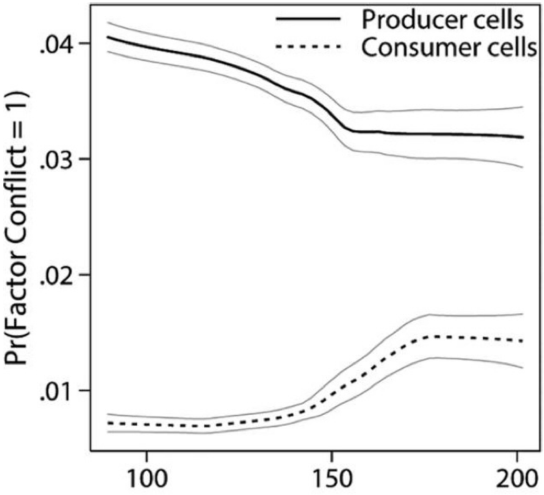
Empirical strategy leverage price and conflict variations at cell i, country l, and year t

$$\text{Factor conflict}_{ilt} = \alpha_i + \sum_{k=0}^{2} \beta^{p}_{ilt-k}\text{PPI}_{ilt-k} + \sum_{k=0}^{2} \beta^{m}_{ilt-k}\text{CPI}_{ilt-k} + \gamma_l \times t + e_{ilt}$$

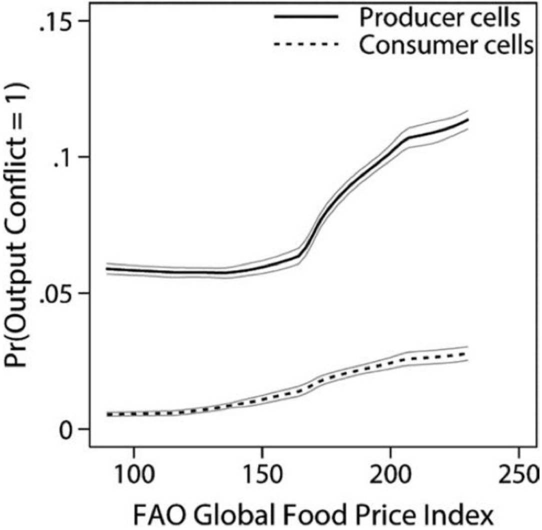
$$\text{Output conflict}_{ilt} = \alpha_i + \sum_{k=0}^{2} \theta^{p}_{ilt-k}\text{PPI}_{ilt-k} + \sum_{k=0}^{2} \theta^{m}_{ilt-k}\text{CPI}_{ilt-k} + \gamma_l \times t + e_{ilt}$$

- Hypotheses: $\beta^p < 0$, but $\beta^m, \theta^p, \theta^m > 0$

Descriptive trends in conflict incidence



(a) Factor conflicts and food price



(b) Output conflicts and food price

Regression results

	Incidence 1(Conflict > 0)		
	(1)	(2)	(3)
PPI	−.0042	$\boxed{−.0046}$	$\boxed{−.0043}$
Conley SE	.001	.001	.001
p-value	.001	.000	.000
Two-way SE	.002	.001	.001
p-value	.007	.001	.002
CPI		$\boxed{.0023}$	$\boxed{.0064}$
Conley SE		.002	.005
p-value		.134	.218
Two-way SE		.001	.006
p-value		.116	.319
PPI impact (%)	−15.4	−17.2	−16.1
CPI impact (%)		8.6	23.7
Wald test: PPI = CPI:			
Conley *p*-value		.000	.040
Two-way *p*-value		.000	.097
Country × year FE	Yes	No	No
Year FE	No	No	Yes
Country × time trend	NA	Yes	Yes
Cell FE	Yes	Yes	Yes
Observations	225,016	204,820	204,820

(a) Factor conflicts and food price

	Incidence 1(Conflict > 0)		
	(1)	(2)	(3)
PPI	.0076	$\boxed{.0095}$	$\boxed{.0090}$
Conley SE	.002	.003	.002
p-value	.000	.000	.000
Two-way SE	.003	.003	.003
p-value	.007	.000	.001
CPI		$\boxed{.0072}$	$\boxed{.0013}$
Conley SE		.002	.006
p-value		.000	.824
Two-way SE		.002	.008
p-value		.000	.864
PPI impact (%)	15.1	18.9	17.8
CPI impact (%)		14.4	2.6
Country × year FE	Yes	No	No
Year FE	No	No	Yes
Country × time trend	NA	Yes	Yes
Cell FE	Yes	Yes	Yes
Observations	173,876	158,270	158,270

(b) Output conflicts and food price

Notable heterogeneity of effects

Effects are weaker in more economically developed areas (factor conflicts)

- Uses nightlight luminosity to proxy economic development (Table 3)
- Coefficient on CPI $\times$ luminosity < 0
- Incidence of conflicts after price shocks less likely in more developed areas
- These areas suffer less income shocks compared to poorer areas

Shocks to food prices have greater impact than cash crop prices

- Cash crops: Non-food crops that does not impact real income of consumers
- Food crop price $\Uparrow \rightarrow$ Conflict $\Uparrow$, Cash crop price $\Uparrow \rightarrow$ Conflict $\Downarrow$

These results suggest household-level economic shocks determine decision to fight

Discussion

Income shocks matter

- Uses price changes that generate opposing income shocks within countries
- Finds opposing trends in the incidence of conflicts within countries
- Rules out the explanation that state's capacity to deter insurgency explains conflicts
- Advances the discussion using much more localized data than in the past

Where to go forward

- Need to locally tailor policy response aimed at minimizing violence
  - Workfare programs for rural areas (keep opportunity cost of joining conflicts high)
  - Buffer stocks or insurance products to shelter consumers from negative income shocks
- Political/historical grievances may still play a role in conflict incidence (Next week)

Cao, Chen (2022): Rebel on the Canal: Disrupted Trade Access and Social Conflict in China, 1650-1911 (student presentation)

Shapiro, Vanden Eynde (2023): Fiscal Incentives for Conflict: Evidence from India's Red Corridor

Q: How does tax revenue structures affect conflict outcomes?

Fiscal revenue structures can shape incentives and prizes for conflict

- Differentiated revenue and expenditure stream in decentralized contexts
- Resource royalties are shared with small number of local governments
- Distribution of such royalties can lead to violence, $\rightarrow$ Govt fail to internalize benefits

This paper

- Utilizes variations in resource royalties across time and types of resources
  - 10-fold increase in iron royalty to states
  - Compare with other resources witout such hikes
- Document rise in state-vs-insurgent conflicts, with aim to take control of these zones

Maoist Conflict in India

Fiscal structure in India

- Central govt sets royalty rates on minerals
- State govt owns resources and receives revenues, but cannot set own rates
- Iron ore royalty to states $\times 10$ since '09: Royalty $\Uparrow$ and Price $\Uparrow$

Mining in India

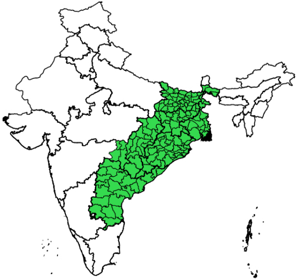
- Largest producers of iron, bauxite, and coal
- Maoist rebels benefit: Take control of mining areas + Extortion from miners

Maoist conflicts (Naxalite)

- Communist uprising against that started as low intensity in 1960s
- Intensified in 2004 following merger of two major factions
- Claimed 5,000+ lives in '07-'13, prevalent in center-east regions (Red Corridor region)

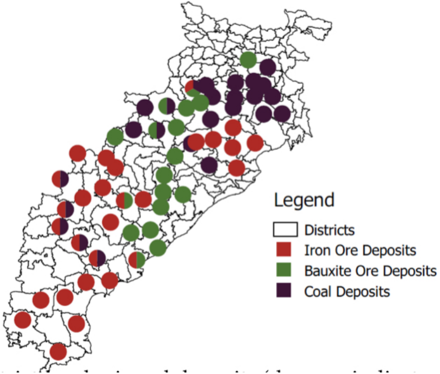
Red Corridor in India

Figure A1: District sample on a map of India



Notes: Districts included in our main sample.

Panel A: Distribution of mineral deposits



Data and Empirics

Data Sources

- Conflict data: South Asia Terrorism Portal (2017-13)
  - Maoist $\rightarrow$ Police, and vice versa, at district level
- Mineral endowments: Geological Survey of India + satellite imagery

Empirical Strategy: Diff-in-diff using introduction of royalty reforms

- OLS with Fixed effects: Compare districts with different iron endowments

$$y_{ist} = \alpha_i + \beta \text{Iron}_{is} \times \text{Post}_t + \gamma \text{Iron}_{is} \times \log \text{Iron price}_t + \mu_{st} + e_{ist}$$

- Main parameter: β (Effect of reforms on iron-producing areas)
- Control for international iron price: Value of iron itself could affect conflicts

Incidence and intensity of violence rises

TABLE 2.—MAIN RESULTS

	Police Attacks on Maoists			Maoist Attacks on Police		
	Dummy	Asinh		Dummy	Asinh	
	(1)	(2)	(3)	(4)	(5)	(6)
Log(Iron Deposit) × Post	0.068***	0.068***	0.074***	0.035*	0.056**	0.054**
	(0.018)	(0.024)	(0.024)	(0.019)	(0.023)	(0.023)
Log(Iron Deposit) × Price (real USD per MT)	0.052**	0.043	0.037	−0.007	−0.004	−0.005
	(0.021)	(0.034)	(0.034)	(0.020)	(0.026)	(0.026)
Log(Bauxite Deposit) × Post			0.016			−0.019
			(0.034)			(0.036)
Log(Coal Deposit) × Post			−0.018			−0.017
			(0.042)			(0.027)
Log(Bauxite Deposit) × Price (real USD per MT)			−0.180			−0.067
			(0.116)			(0.115)
Log(Coal Deposit) × Price (real USD per MT)			0.070			0.053
			(0.057)			(0.060)
Mean DV	0.140	0.175	0.175	0.116	0.136	0.136
Observations	1440	1440	1440	1440	1440	1440
Clusters	144	144	144	144	144	144

Regressions at the district-half-year level (2007–2011). All regressions include district fixed effects, and state × time fixed effects. Standard errors are clustered at the district level and presented in parentheses: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.1$.

- Mean of log(Iron): 1.7 $\rightarrow$ What happens by moving from no-iron to mean-iron area?
- Incidence: Police-on-Maoist violence 12 p.p (=0.068 $\times$ 1.7)$\Uparrow$
- Intensity: Increase by 12% and 10% each (=0.056 $\times$ 1.7)

Red Corridor in India

TABLE 3.—ILLEGAL MINING

	Excess Mining (0-1) (1)	Excess Mining 50% threshold (0-1) (2)	Truck Activity (0-1) (3)
Iron Mine × (2009–2011)	0.132^{*}	0.127^{**}	-0.141^{*}
	(0.068)	(0.063)	(0.083)
Iron Mine × (2012–2013)	0.156^{**}	0.132^{**}	−0.088
	(0.068)	(0.062)	(0.128)
Iron Mine × (2009–2011) × Expired			0.218^{*}
			(0.124)
Iron Mine × (2012–2013) × Expired			−0.044
			(0.175)
Mean DV	0.276	0.228	0.701
Observations	558	558	558
Clusters	82	82	82

Annual mine-level data between 2007 and 2013. Truck activity and area measurements are based on satellite imagery analyzed for the purpose of this study. The illegal mining proxies are based on comparing satellite nformation with the legal area and expiration date from the India *Directory of Mining Leases*. The excess mining indicator is equal to 1 when the measured area exceeds the legal area (by more than 50% in column :). Control minerals include bauxite, manganese, and chromite. Regressions are mine, state × time, and (in column 3) expired status × time fixed effects. Standard errors are clustered at the mine level and presented n parentheses: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.1$.

- Increase in illegal mining by 15 p.p
- State's counterinsurgency ↑ for rents or miners help Maoists to keep illegal activities

Discussion

What drives violence (or not)

- Effect fades out: State seeems to have established control
- Not driven by resource prices: No statistical relationship with conflicts
- This effect not shown with other types of resources
- Changes in incentives: Reflected in illegal mining activities

Takeaways

- Fiscal revenue structure can explain conflicts
- Driven by increased security efforts or miners funding conflict to avoid royalties
- There are cases where bandits take over, especially in state absence
  (See Sanchez de la Sierra 2020, to be covered later in semester)

Blattman, Annan (2016): Can Employment Reduce Lawlessness and Rebellion?

Q: Does incentives to individuals affect conflict participation

Many fragile states provide work to deter interest in conflicts and illicit activities

- Aim to stimulate employment by providing capital and training
- Decrease incentives for illegal work
- Provide income to integrate soldiers back into economy

However, ex-fighters (High-risk men) are different

- Comparative advantage in violence over human/social capital
- Generally show lack of interest in existing rehabilitation programs

This paper experimentally assesses the effect of providing capital on conflict participation

- Experiments effects of providing agricultural training and capital inputs to ex-soldiers
- Highlights that long-run economic incentives can deter conflict participation

Liberian Civil wars

History

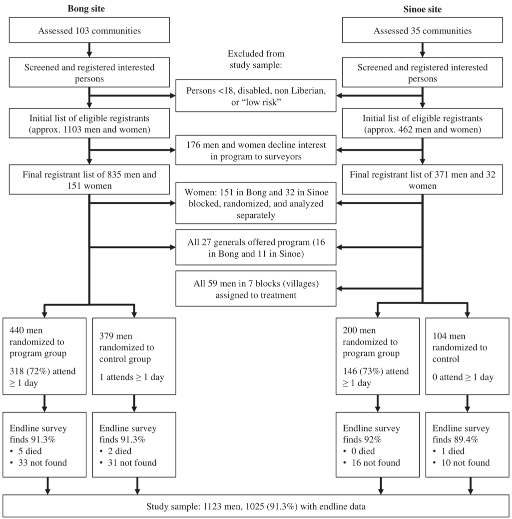
- Civil wars in 1989-1996 and 1999-2003
- Killed 10% of population, with many young men recruited to fight
- By 2008: Ex-soldiers engaged in illicit activities while clustered in remote ‘hotspots’

Experiment

- Takes place in 2009 in two regions (Bong and Sinoe)
- Trains agricultural skills and social skills during program
- After graduation, participants provided with farming tools worth around $125
- Note that some were given cash (implementation mistakes)

Research Design

FIGURE 1. Sample Recruitment, Selection, Randomization, and Data Collection



Empirical Strategy

- Program assignment randomized: OLS regression between treated vs control group
- Randomized within blocks defined by training site, rank, community

TABLE 1. Baseline Summary Statistics and Test of Randomization Balance

		Test of Balance ($n = 1025$)	
Baseline Covariate	Control Mean (1)	Treatment Difference (2)	p Value (3)
Age	30.5	−0.617	0.256
Lives with Spouse/Partner	0.722	−0.035	0.215
Number of Children	2.5	−0.207	0.148
Disabled, Injured, or Ill	0.349	−0.011	0.664
Years of Schooling	5.68	0.210	0.364
Said Would Attend if Selected	0.984	0.002	0.616
Durable Assets Index (z score)	−0.04	−0.006	0.909
Stock of Savings (USD)	44.2	−13.066	0.025
Debt Stock (USD)	7.2	−0.573	0.587
Agricultural Experience Index (z score)	0.09	−0.062	0.390
Aggressive Behaviors Index (0–12)	1.18	0.125	0.248
Main Income: Illicit Resources	0.228	0.001	0.955
Main Income: Legal Nonfarm Work	0.292	−0.029	0.362
Very Interested in Farming	0.863	−0.012	0.561
Can Access 10 Acres Farmland	0.90	0.03	0.21
Ex-combatant	0.727	−0.006	0.826
Months in a Faction	23.8	5.731	0.002
Ex-commander Relations Index (z score)	0.04	−0.085	0.171
Patience Index (0–4)	2.97	0.010	0.878
Risk Affinity Index (0–3)	0.33	0.037	0.370

Notes: Columns (2) and (3) report results of an OLS regression of each covariate on the treatment indicator and block fixed effects. Standard errors are clustered by village. USD variables are censored at the 99th percentile.

Shift away from illicit activities to agriculture

TABLE 3. Program Impacts on Occupational Choice and Income

Outcome	Control Mean (1)	ITT Coeff. (2)	ITT Std. Err. (3)	TOT Coeff. (4)	TOT Std. Err. (5)
		Treatment Effect Estimates ($n = 1025$)			
Agricultural Engagement:					
Raising Crops/Animals†	0.61	0.118	[0.030]***	0.155	[0.036]***
Acres under Cultivation	4.43	1.556	[2.146]	2.037	[2.573]
Thinks Farming is a Good Living	0.95	0.008	[0.016]	0.010	[0.019]
Interested in Farming	0.78	0.090	[0.029]***	0.118	[0.035]***
Interested in Raising Animals	0.90	0.049	[0.019]**	0.064	[0.023]***
Hours Worked/Week, Past Month	49.33	0.978	[2.357]	1.278	[2.824]
Illicit Resource Extraction	15.57	−2.829	[1.350]**	−3.697	[1.593]**
Logging	2.79	−0.926	[0.649]	−1.210	[0.773]
Mining	10.53	−1.356	[1.140]	−1.772	[1.362]
Rubber Tapping	2.25	−0.547	[0.573]	−0.715	[0.682]
Farming and Animal-raising	11.91	3.131	[1.180]***	4.090	[1.415]***
Farming	10.45	2.620	[1.037]**	3.423	[1.242]***
Animal-raising	1.46	0.511	[0.508]	0.667	[0.609]
Contract Agricultural Labor	1.82	−0.116	[0.320]	−0.152	[0.383]
Palm, Coconut, Sugar Cutting	0.34	0.264	[0.343]	0.345	[0.413]
Hunting	1.18	0.215	[0.334]	0.281	[0.401]
Non-farm Labor and Business	18.16	−0.170	[2.055]	−0.222	[2.464]
Other Activities	0.36	0.483	[0.571]	0.632	[0.682]

- More interest in agriculuture, but not complete shift from illicit activities

Decreased interst in violence following the program

TABLE 4. Program Impacts on Mercenary Recruitment Proxies

Outcome	Control Mean (1)	TOT Estimate Coeff. (2)	TOT Estimate SE (3)
All Recruitment Interest/Actions (*z* score)	0.09	−0.204	[0.079]***
Direct Recruitment Proxies (0–12)	0.94	−0.239	[0.118]**
Talked to a Commander in Last 3 Months	0.45	−0.108	[0.044]**
Would Go if Called to Fight for Tribe	0.05	−0.015	[0.013]
Has Been Approached about Going to CI	0.07	0.001	[0.021]
Would Go to CI for $250	0.01	−0.006	[0.010]
Would Go to CI for $500	0.03	−0.009	[0.012]
Would Go to CI for $1000	0.08	−0.041	[0.019]**
Will Move Towards CI Border Area	0.10	−0.022	[0.024]
Invited to Secret Meeting on Going to CI	0.04	0.004	[0.016]
Attended Secret Meeting on Going to CI	0.03	−0.013	[0.011]
Was Promised Money to Go to CI	0.03	0.001	[0.014]
Willing to Fight if War Breaks Out in CI	0.04	−0.018	[0.015]
Has Plans to Go to CI in the Next Month	0.01	−0.012	[0.009]
Indirect Recruitment Proxies (0–4)	1.48	−0.158	[0.076]**
Talks about the CI Violence with Friends	0.68	−0.046	[0.041]
Has a Partisan Preference in CI	0.66	−0.117	[0.041]***
Knows People Who Went to CI to Fight	0.10	−0.021	[0.019]
Knows People Given Money to Go to CI	0.04	0.026	[0.016]

Other notable outcomes and discussions

Other findings

- Limited effect on networking and integration
  - Combintation of economic interventions and other social interventions required?
- Impacts of training alone without capital lower
  - Provision of *long-run* economic incentives required for effective rehabilitation

Takeaway

- Individual's decision to participate in illicit activities respond to economic margins
- However, these should come with long-run horizons and may not affect integration
- Economic and behavioral treatment suggested for effective rehabiliation (also generalizable to developed countries)

Berman et al (2011):
Do Working Men Rebel? Insurgency and Unemployment in Afghanistan, Iraq, and the Philippines [student presentation]

Takeaways and remaining questions

Economic incentives do determine decision to fight

- Resource endowments shape returns from participating in war vs peaceful activities
- High opportunity costs deter participating in conflicts
- Individual incentives matter: Largely comes from benefits of labor market vs conflicts

Unanswered questions

- How to design incentives to deter interest in conflicts?
- Some institutions prevent incentives from spilling into conflicts: What? How?
- Other determinants of conflict: Cultural factors, misperceptions from misinformation, among other things